



Lab 08: Pub/Sub Qwik Start - Command Line

Google Cloud Associate Cloud Engineer | GSP095

Overview

Performed all Pub/Sub operations using gcloud commands in Cloud Shell. Created and deleted topics and subscriptions, published multiple messages, and learned how the pull command works with and without flags.

Objectives

- Create, list, and delete Pub/Sub topics
- Create, list, and delete Pub/Sub subscriptions
- Publish multiple messages to a topic
- Pull messages using the pull command
- Pull all messages at once using the limit flag

Task 1 - Pub/Sub Topics

```
gcloud pubsub topics create myTopic
gcloud pubsub topics create Test1
gcloud pubsub topics create Test2
gcloud pubsub topics list
gcloud pubsub topics delete Test1
gcloud pubsub topics delete Test2
```

```

student_01_afdc55dd1197@cloudshell:~ (qwiklabs-gcp-00-88a12dca3ac6)$ gcloud pubsub
topics list
---
messageStoragePolicy:
  allowedPersistenceRegions:
    - us-central1
name: projects/qwiklabs-gcp-00-88a12dca3ac6/topics/Test2
---
messageStoragePolicy:
  allowedPersistenceRegions:
    - us-central1
name: projects/qwiklabs-gcp-00-88a12dca3ac6/topics/Test1
---
messageStoragePolicy:
  allowedPersistenceRegions:
    - us-central1
name: projects/qwiklabs-gcp-00-88a12dca3ac6/topics/myTopic
student_01_afdc55dd1197@cloudshell:~ (qwiklabs-gcp-00-88a12dca3ac6)$ gcloud pubsub
topics delete Test1
Deleted topic [projects/qwiklabs-gcp-00-88a12dca3ac6/topics/Test1].
student_01_afdc55dd1197@cloudshell:~ (qwiklabs-gcp-00-88a12dca3ac6)$ gcloud pubsub
topics delete Test2
Deleted topic [projects/qwiklabs-gcp-00-88a12dca3ac6/topics/Test2].
student_01_afdc55dd1197@cloudshell:~ (qwiklabs-gcp-00-88a12dca3ac6)$ gcloud pubsub
topics list
---
messageStoragePolicy:
  allowedPersistenceRegions:
    - us-central1
name: projects/qwiklabs-gcp-00-88a12dca3ac6/topics/myTopic
student_01_afdc55dd1197@cloudshell:~ (qwiklabs-gcp-00-88a12dca3ac6)$ █

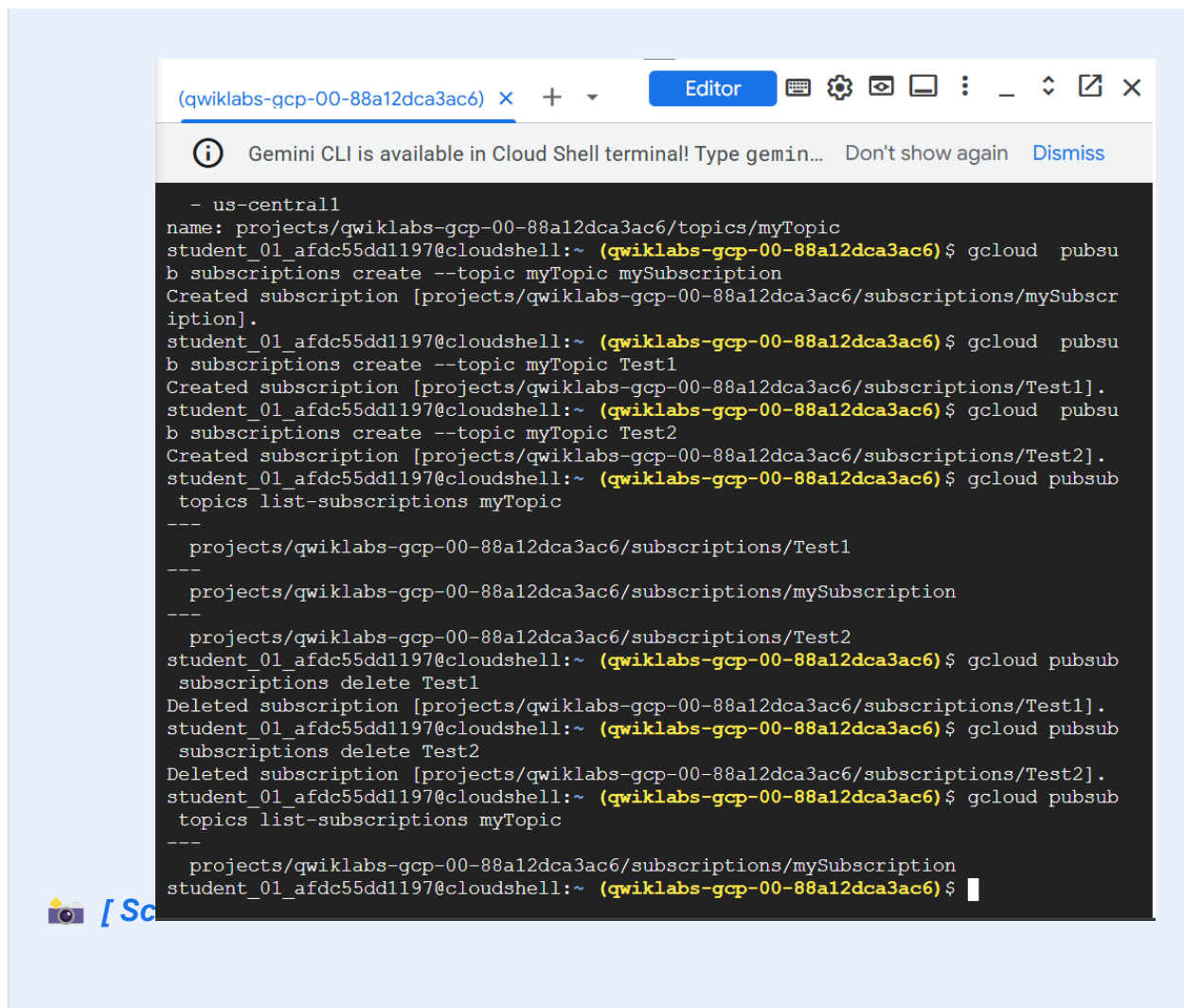
```

Task 2 - Pub/Sub Subscriptions

```

gcloud pubsub subscriptions create --topic myTopic mySubscription
gcloud pubsub subscriptions create --topic myTopic Test1
gcloud pubsub subscriptions create --topic myTopic Test2
gcloud pubsub topics list-subscriptions myTopic
gcloud pubsub subscriptions delete Test1
gcloud pubsub subscriptions delete Test2

```



The screenshot shows a Cloud Shell terminal window with the title bar "(qwiklabs-gcp-00-88a12dca3ac6) x + ▾ Editor". A notification bar at the top states "Gemini CLI is available in Cloud Shell terminal! Type gemin... Don't show again Dismiss". The terminal output shows the following commands and results:

```
- us-central1
name: projects/qwiklabs-gcp-00-88a12dca3ac6/topics/myTopic
student_01_afdc55dd1197@cloudshell:~ (qwiklabs-gcp-00-88a12dca3ac6) $ gcloud pubsub
subscriptions create --topic myTopic mySubscription
Created subscription [projects/qwiklabs-gcp-00-88a12dca3ac6/subscriptions/mySubscr
ption].
student_01_afdc55dd1197@cloudshell:~ (qwiklabs-gcp-00-88a12dca3ac6) $ gcloud pubsub
subscriptions create --topic myTopic Test1
Created subscription [projects/qwiklabs-gcp-00-88a12dca3ac6/subscriptions/Test1].
student_01_afdc55dd1197@cloudshell:~ (qwiklabs-gcp-00-88a12dca3ac6) $ gcloud pubsub
subscriptions create --topic myTopic Test2
Created subscription [projects/qwiklabs-gcp-00-88a12dca3ac6/subscriptions/Test2].
student_01_afdc55dd1197@cloudshell:~ (qwiklabs-gcp-00-88a12dca3ac6) $ gcloud pubsub
topics list-subscriptions myTopic
---
projects/qwiklabs-gcp-00-88a12dca3ac6/subscriptions/Test1
---
projects/qwiklabs-gcp-00-88a12dca3ac6/subscriptions/mySubscription
---
projects/qwiklabs-gcp-00-88a12dca3ac6/subscriptions/Test2
student_01_afdc55dd1197@cloudshell:~ (qwiklabs-gcp-00-88a12dca3ac6) $ gcloud pubsub
subscriptions delete Test1
Deleted subscription [projects/qwiklabs-gcp-00-88a12dca3ac6/subscriptions/Test1].
student_01_afdc55dd1197@cloudshell:~ (qwiklabs-gcp-00-88a12dca3ac6) $ gcloud pubsub
subscriptions delete Test2
Deleted subscription [projects/qwiklabs-gcp-00-88a12dca3ac6/subscriptions/Test2].
student_01_afdc55dd1197@cloudshell:~ (qwiklabs-gcp-00-88a12dca3ac6) $ gcloud pubsub
topics list-subscriptions myTopic
---
projects/qwiklabs-gcp-00-88a12dca3ac6/subscriptions/mySubscription
student_01_afdc55dd1197@cloudshell:~ (qwiklabs-gcp-00-88a12dca3ac6) $
```

Task 3 - Publish and Pull a Single Message

```
gcloud pubsub topics publish myTopic --message "Hello"
gcloud pubsub topics publish myTopic --message "Publisher's name is [YOUR NAME]"
gcloud pubsub topics publish myTopic --message "Publisher likes to eat [FOOD]"
gcloud pubsub topics publish myTopic --message "Publisher thinks Pub/Sub is
awesome"
gcloud pubsub subscriptions pull mySubscription --auto-ack
```

Note: Pull without flags only returns one message at a time. Once pulled a message cannot be pulled again from that subscription.

```

Publisher thinks Pub/Sub is awesome"
messageIds:
- '19623898912982428'
student_01_afdc55dd1197@cloudshell:~ (qwiklabs-gcp-00-88a12dca3ac6) $ gcloud pubsub subscriptions pull
auto-ack
DATA: Publisher likes to eat pink lady apples
MESSAGE ID: 19623434049561091
ORDERING_KEY:
ATTRIBUTES:
DELIVERY_ATTEMPT:
ACK STATUS: SUCCESS
student_01_afdc55dd1197@cloudshell:~ (qwiklabs-gcp-00-88a12dca3ac6) $ 

```

Task 4 - Pull All Messages Using Limit Flag

```

gcloud pubsub topics publish myTopic --message "Publisher is starting to get the
hang of Pub/Sub"
gcloud pubsub topics publish myTopic --message "Publisher wonders if all messages
will be pulled"
gcloud pubsub topics publish myTopic --message "Publisher will have to test to find
out"
gcloud pubsub subscriptions pull mySubscription --limit=3

```

```

ACK STATUS: SUCCESS
student_01_afdc55dd1197@cloudshell:~ (qwiklabs-gcp-00-88a12dca3ac6) $ gcloud pubsub topics publish myTopic --message "Pu
blisher is starting to get the hang of Pub/Sub"
messageIds:
- '19624230372878886'
student_01_afdc55dd1197@cloudshell:~ (qwiklabs-gcp-00-88a12dca3ac6) $ gcloud pubsub topics publish myTopic --message "Pu
blisher wonders if all messages will be pulled"
messageIds:
- '19625337776456202'
student_01_afdc55dd1197@cloudshell:~ (qwiklabs-gcp-00-88a12dca3ac6) $ gcloud pubsub topics publish myTopic --message "Pu
blisher will have to test to find out"
messageIds:
- '19613573467066391'
student_01_afdc55dd1197@cloudshell:~ (qwiklabs-gcp-00-88a12dca3ac6) $ gcloud pubsub subscriptions pull mySubscription --
limit=3
DATA: Publisher will have to test to find out
MESSAGE ID: 19613573467066391
ORDERING_KEY:
ATTRIBUTES:
DELIVERY_ATTEMPT:
ACK ID: BhYsXUZIUTcZCGhRdk9eIz81IChFEQIATwIoXXkwVzy5BWHMPg0Zcn1hdGMPFQFQR1p9C1IRDWJcTkQHSZ_4kJ1XV0tbGgcDR1Z5XF8dCmhaWnY
OUCXtwfz-09C3ARs-fcGfvbstLb39jbI1ZiI9XhJLLD5-MD9FQV5AEkw7CkRJUytdCypYEU4EISE-MD5FU0RQ

DATA: Publisher wonders if all messages will be pulled
MESSAGE ID: 19625337776456202
ORDERING_KEY:
ATTRIBUTES:
DELIVERY_ATTEMPT:
ACK ID: BhYsXUZIUTcZCGhRdk9eIz81IChFEQIATwIoXXkwVzy5BWHMPg0Zcn1hdGMPFQFQR1p9C1IRDWJcTkQHSZ_4kJ1XV0tbGgcAqFB9WFwcC2xZWnc
HUYXtwfz-09C3ARs-fcGfvbstLb39jbI1ZiI9XhJLLD5-MD9FQV5AEkw7CkRJUytdCypYEU4EISE-MD5FU0RQ
student_01_afdc55dd1197@cloudshell:~ (qwiklabs-gcp-00-88a12dca3ac6) $ 

```

What I Learned

- Pull without flags returns only one message even if multiple exist

- Once a message is pulled from a subscription it cannot be pulled again
- The --limit flag sets how many messages to pull in one request
- The --auto-ack flag automatically acknowledges messages after pulling
- Topics and subscriptions can be created, listed, and deleted via CLI

Resources

GCP Pub/Sub Docs: <https://cloud.google.com/pubsub/docs>

Lab ID: GSP095